

Pandas Complete Notes

What is Pandas?

Pandas is a powerful Python library used for:

- Data analysis
- Data cleaning
- Data manipulation
- Handling structured data (like Excel, CSV, SQL tables)

It works mainly with two data structures:

1. **Series** → 1D (like a column)
 2. **DataFrame** → 2D (like a table/spreadsheet)
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Installing Pandas

`pip install pandas`

Import it:

`import pandas as pd`

Core Data Structures

1 Series

`import pandas as pd`

`s = pd.Series([10, 20, 30])`

`print(s)`

Create Series with custom index

```
s = pd.Series([10, 20, 30], index=['a','b','c'])
```

2 DataFrame

```
data = {  
    "Name": ["John", "Anna", "Peter"],  
    "Age": [28, 24, 35]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```



Reading Data

Function	Description
<code>pd.read_csv()</code>	Read CSV file
<code>pd.read_excel()</code>	Read Excel file
<code>pd.read_json()</code>	Read JSON
<code>pd.read_sql()</code>	Read SQL database
<code>pd.read_html()</code>	Read tables from HTML

Example

```
df = pd.read_csv("data.csv")
```



Writing Data

Function	Description
<code>df.to_csv()</code>	Save to CSV
<code>df.to_excel()</code>	Save to Excel
<code>df.to_json()</code>	Save to JSON
<code>df.to_sql()</code>	Save to SQL



Viewing Data

Function	Explanation
<code>df.head()</code>	First 5 rows
<code>df.tail()</code>	Last 5 rows
<code>df.info()</code>	Summary of dataframe

`df.describe()` Statistical summary

`df.shape` (rows, columns)

`df.columns` Column names

`df.index` Row index

Selecting Data

Select Column

```
df["Name"]
```

```
df[["Name", "Age"]]
```

Select Rows

```
df.loc[0] # by label
```

```
df.iloc[0] # by position
```

Filter Data

```
df[df["Age"] > 25]
```

Adding & Modifying Columns

```
df["Salary"] = [50000, 60000, 70000]
```

```
df["Age"] = df["Age"] + 1
```

Handling Missing Data

Function	Description
<code>df.isnull()</code>	Check missing
<code>df.notnull()</code>	Not missing
<code>df.dropna()</code>	Remove missing
<code>df.fillna()</code>	Replace missing

Example:

```
df.fillna(0)
```



Sorting

```
df.sort_values("Age")
```

```
df.sort_index()
```



Aggregation Functions

Function	Explanation
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<code>sum()</code>	Total
--------------------	-------

<code>mean()</code>	Average
---------------------	---------

<code>median()</code>	Middle value
-----------------------	--------------

<code>min()</code>	Minimum
--------------------	---------

<code>max()</code>	Maximum
--------------------	---------

<code>count()</code>	Count values
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<code>std()</code>	Standard deviation
--------------------	--------------------

<code>var()</code>	Variance
--------------------	----------

Example:

```
df["Age"].mean()
```

GroupBy (Very Important)

```
df.groupby("Department")["Salary"].mean()
```

Common group functions:

- `sum()`
 - `mean()`
 - `count()`
 - `max()`
 - `min()`
-

Merge, Join & Concat

Merge

```
pd.merge(df1, df2, on="id")
```

Types:

- inner
- outer
- left
- right

Concat

```
pd.concat([df1, df2])
```

Join

```
df1.join(df2)
```



Apply Functions

```
df["Age"].apply(lambda x: x+5)
```



Dropping Data

```
df.drop("Age", axis=1)    # drop column
```

```
df.drop(0, axis=0)       # drop row
```



Rename Columns

```
df.rename(columns={"Age": "Years"})
```



Unique & Value Counts

```
df["Department"].unique()
```

```
df["Department"].value_counts()
```



Replace Values

```
df.replace("Male", "M")
```



Indexing & Setting Index

```
df.set_index("Name")
```

```
df.reset_index()
```



Iterating Through Data

for index, row in df.iterrows():

```
    print(row["Name"])
```



Pivot Tables

```
pd.pivot_table(df, values="Salary", index="Department", aggfunc="mean")
```



Working with Dates

```
df["Date"] = pd.to_datetime(df["Date"])
```

```
df["Year"] = df["Date"].dt.year
```



Basic Plotting

```
df["Age"].plot()
```

String Functions

```
df["Name"].str.lower()
```

```
df["Name"].str.upper()
```

```
df["Name"].str.contains("a")
```

Mathematical Operations

```
df["Age"] + 5
```

```
df["Salary"] * 2
```

Important Attributes

Attribute	Description
<code>df.values</code>	Numpy array
<code>df.dtypes</code>	Data types
<code>df.ndim</code>	Number of dimensions
<code>df.size</code>	Total elements
<code>df.T</code>	Transpose



Complete List of Common Pandas Functions

Below is a categorized list of **most used pandas functions**:

Creation

- `Series()`
- `DataFrame()`

Reading

- `read_csv()`
- `read_excel()`
- `read_json()`
- `read_sql()`
- `read_html()`

Writing

- `to_csv()`
- `to_excel()`
- `to_json()`
- `to_sql()`

Selection

- `loc`
- `iloc`
- `at`
- `iat`

Cleaning

- `drop()`
- `dropna()`
- `fillna()`
- `replace()`
- `rename()`

Aggregation

- `sum()`
- `mean()`
- `median()`
- `mode()`
- `count()`
- `std()`
- `var()`

Grouping

- `groupby()`
- `agg()`

Combining

- `merge()`
- `join()`
- `concat()`

Sorting

- `sort_values()`
- `sort_index()`

Reshaping

- `pivot()`
- `pivot_table()`
- `melt()`
- `stack()`
- `unstack()`

Apply

- `apply()`
- `map()`
- `applymap()`

Beginner Tips

✓ Always check:

`df.head()`

`df.info()`

`df.describe()`

✓ Handle missing data early

✓ Use `groupby()` for analysis

✓ Use `merge()` carefully (know join type)

1.

